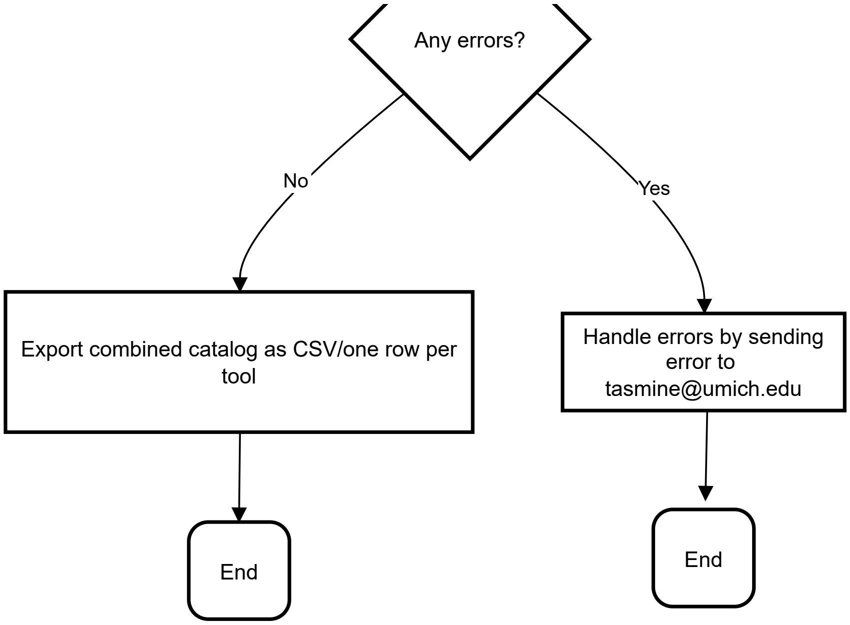
Bacterial Data Analysis Tools Catalog Pipeline

Parallel API queries merge into unified, enriched catalog



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1. Functions and Modules
    a. **Search Term Handling**
        i. **make_tag_pattern** function takes a vector of tag keywords and makes a single string separated by "|" in order to make regex expression matching with the tags simpler
    b. **Tool Classification**
        i. **classify_tool** groups tags into the following categories (Visualization tools, Sequence analysis, Phylogenetic and evolutionary analysis, Comparative genomics, Metagenomics and community analysis, Specialized analysis)
    c. **Receive package/tool list and add to dataframe**
        i. **tool_list_to_df** takes a list of tool metadata entries, replaces any NULL fields with empty strings, converts each entry to a data frame, and then combines all of those into one big data frame.
2. **Query Steps & Functions**
    a. Query_cran_by_tags
        i. Extract raw metadata
        ii. Get_citation_from_doi
        iii. Get_citation_count_cached
    b. Query_bioc_by_tags
        i. Extract raw metadata
        ii. Get_citation_from_doi
        iii. Get_citation_count_cached
    c. Search_github_tools using a GitHub token
        i. **Extract raw metadata** from https://api.github.com/search/repositories for the following field domains:
            1. tibble(
            2. topic = topic,
            3. name = dat$items$name,
            4. owner = dat$items$owner.login,
            5. author = dat$items$owner.login,
            6. summary = dat$items$description,
            7. doi = NA_character_,
            8. citation = NA_character_,
            9. apa_citation= NA_character_,
            10. language = dat$items$language,
            11. release_date= dat$items$created_at,
            12. docs = dat$items$html_url,
            13. tutorials = NA_character_,
            14. license = license_names,
            15. stars = dat$items$stargazers_count,
            16. source = "GitHub”)
        ii. Limit github query by month/year to decrease limits

3. **Merge metadata** fields across CRAN/Bioconductor/ GitHub ensuring that formatting is consistent across tools in the final merged csv file
4. **Enrich the metadata** by combining information from any duplicate tools/packages
5. **Validate and de-duplicate tools** by removing duplicate rows from the dataframe
6. **Evaluate errors**
    a. If there are errors:
        i. remove them from database fields and replace them with blank space and create csv output file
        ii. Email tasmine@umich.edu
    b. If there are no errors:
        i. Create csv output file

Project Goals and Milestones

Goals:

- Develop an automated and unified software catalog for bacterial data analysis tools.
- Enable tool discovery, metadata enrichment, classification, and rigorous validation from multiple repositories (CRAN, Bioconductor, GitHub).
- Export a de-duplicated, comprehensive CSV catalog for researcher use.

Milestones:

1. Set up repository and push documentation.
2. Implement functional modules (search/aggregation, classification, enrichment, export).
3. Test with representative and full-scale datasets.
4. Complete error handling and validation routines.
5. Release user documentation and tutorial.
6. Finalize and demonstrate the pipeline with newest data.